



Accredited with Grade A+ by NAAC, GoI

ACADEMIC REGULATIONS & SYLLABI



Faculty of Management Studies
Bachelor of Business Administration (BBA) – Business
Analytics
Batch: 2023-26



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY - CHARUSAT

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Education Campus – Changa, (ECC), hitherto a conglomerate of institutes of professional education in Engineering, Pharmacy, Computer Applications, Management, Applied Sciences, Physiotherapy and Nursing, is one of the choicest destinations by students. It has been transformed into Charotar University of Science and Technology (CHARUSAT) through an Act by Government of Gujarat. CHARUSAT is permitted to grant degrees under Section-22 of UGC – Government of India.

The journey of CHARUSAT started in the year 2000, with only 240 Students, 4 Programmes, one Institute and an investment of about Rs. 3 Crores (INR 30 million). At present there are nine different institutes falling under ambit of six different faculties. The programmes offered by these faculties range from undergraduate (UG) to Ph.D degrees. These faculties, in all offer 74 different programmes. A quick glimpse is as under:

Faculty	Institute	Programmes Offered
Faculty of Technology and Engineering	Chandubhai S. Patel Institute of Technology	B.Tech M.Tech Ph.D
Faculty of Pharmacy	Ramanbhai Patel College of Pharmacy	B.Pharm M.Pharm Ph.D PGDCT / PGDPT
Faculty of Management Studies	Indukaka Ipcowala Institute of Management	BBA MBA Ph.D
Faculty of Computer Applications	Smt. Chandaben Mohanbhai Patel Institute of Computer Applications.	BCA MCA M.Sc IT Ph.D BCA
Faculty of Sciences	P.D. Patel Institute of Applied Sciences	B.Sc M.Sc Ph.D
Faculty of Medical Sciences	Ashok and Rita Institute of Physiotherapy	B.PT M.PT Ph.D

	Manikaka Topawala Institute of Nursing	B.Sc (Nursing) P.B. B.Sc. Nursing GNM M.Sc. Ph.D
	Bapubhai Desaibhai Patel Institute of Paramedical Sciences	BMIT BSMT BOTAT BOP MSc. MIT MSMLT MSOTAT PGDMLT PGDCH Ph. D

The development and growth of the institutes have already led to an investment of over Rs.150 Crores (INR 1500 Million). The future outlay is planned with an estimate of Rs.250 Crores (INR 2500 Million).

The University is characterized by state-of-the-art infrastructural facilities, innovative teaching methods and highly learned faculty members. The University Campus sprawls over 125 acres of land and is Wi-Fi enabled. It is also recognized as the Greenest Campus of Gujarat.

CHARUSAT is privileged to have 360 core faculty members, educated and trained in IITs, IIMs and leading Indian Universities, and with long exposure to industry. It is also proud of its past students who are employed in prestigious national and multinational corporations.

From one college to the level of a forward-looking University, CHARUSAT has the vision of entering the club of premier Universities initially in the country and then globally. **High Moral Values like Honesty, Integrity and Transparency** which has been the foundation of ECC continues to anchor the functioning of CHARUSAT. Banking on the world class infrastructure and highly qualified and competent faculty, the University is expected to be catapulted into top 20 Universities in the coming five years. In order to align with the global requirements, the University has collaborated with internationally reputed organizations like Pennsylvania State University – USA, University at Alabama at Birmingham – USA, Northwick Park Institute –UK, ISRO, BARC, etc.

CHARUSAT has designed curricula for all its programmes in line with the current international practices and emerging requirements. Industrial Visits, Study Tours, Expert Lectures and Interactive IT enabled Teaching Practice form an integral part of the unique CHARUSAT pedagogy.

The programmes are credit-based and have continuous evaluation as an important feature. The pedagogy is student-centered, augurs well for self-learning and motivation for enquiry and research, and contains innumerable unique features like:

- Participatory and interactive discussion-based classes.
- Sessions by visiting faculty members drawn from leading academic institutions and industry.
- Regular weekly seminars.
- Distinguished lecture series.
- Practical, field-based projects and assignments.
- Summer training in leading organizations under faculty supervision in relevant programmes.
- Industrial tours and visits.
- Extensive use of technology for learning.
- Final Placement through campus interviews.

Exploration in the field of knowledge through research and development and comprehensive industrial linkages will be a hallmark of the University, which will mould the students for global assignments through technology-based knowledge and critical skills.

The evaluation of the student is based on grading system. A student has to pursue his/her programme with diligence for scoring a good Cumulative Grade Point Average (CGPA) and for succeeding in the chosen profession and life.

CHARUSAT welcomes you for a Bright Future.....



CHAROTAR UNIVERSITY OF SCIENCE AND
TECHNOLOGY
Faculty of Management Studies

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ACADEMIC REGULATIONS

BBA Programme

Bachelor of Business Administration (BBA) Programme
Business Analytics

Academic Year: 2024-25

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
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CHARUSAT

FACULTY OF MANAGEMENT STUDIES

ACADEMIC REGULATIONS

Bachelor of Business Administration (BBA) - BA

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a specified load of course work in the chosen subject of specialization and also complete a project/dissertation if any.

2. Duration of Programme

2.1 The Bachelor of Business Administration (BBA) – *Business Analytics* programme of Charotar University of Science and Technology (CHARUSAT) is a **three year full-time** under-graduate programme, leading to the award of the degree of Bachelor of Business Administration (BBA).

2.2 The span of the programme is six years (as decided by the university from time to time) from the date of registration in the programme.

3. Eligibility and mode of admissions

Candidate should have passed 12th standard (or equivalent) examination (any stream) with English as a subject is eligible for admission to the BBA – Business Analytics Programme and according to the regulations for admission decided by CHARUSAT from time to time.

4. Programme structure and Credits

4.1 *A student admitted to a program should study the course and earn credits specified in the course structure.* The details of programme structure, credit requirements, areas of specialisation proposed to be offered, etc. are presented at Appendix – I.

4.2 Programme Outcomes (PO)

Graduates will be able to...

PO1 - Analyse contemporary issues and apply management knowledge for resolution.

PO2 - Practice effective communication and interpersonal skills.

PO3 - Understand Global Perspectives and exhibit Cross-Cultural sensitivity.

PO4 - Demonstrate Attitude for continuous learning and development.

5. Attendance

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will require to be condoned by the Dean/Principal.

5.2 Student attendance in every course should be 80%.

6. Course Evaluation

6.1 The performance of every student in each course will be evaluated as follows:

6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these, for 70% of the marks for the course.

6.2 University Examination

6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation or a combination of these.

6.2.2 In order to earn the credit in a course, a student has to obtain a grade other than FF.

6.3 Performance at Internal Evaluation Components and University Examination

If a student secures minimum passing marks of 40% in the University examinations in any course but fails to obtain the minimum passing total percentage of 50%, he/she has to repeat the university examination in the course.

Minimum percentage marks in University Exam for pass in any course	Minimum total percentage marks (i.e. internal+ University) for pass in any course
40%	50%

7. Grading

7.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Grading Scheme:

Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

7.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses in the semester

(ii) CGPA = $\frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move to the second academic year if his/her CGPA is less than 3 at the end of the first academic year.

(iv) in addition to above, a student has to comply with the requirements of the regulatory bodies, wherever such requirements exist.

(v) A student will have a maximum of four chances* after first appearing in that examination to clear that course, subject to the restriction on the span period stipulated in clause 2.2 above.

*(*Whenever the university conducts the examinations of that course, it will be considered as a chance, irrespective of whether the student appears for the examination or not.)*

8. Awards of Bachelor Degree

8.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

8.1.1 He should have earned at least minimum required credits as prescribed in course structure.

8.2 Any student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree.

9. Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

First Class with Distinction	= CGPA of. 7.50 and above
First Class	= CGPA of 6.00 to 7.49
Second Class	= CGPA of 5.00 to 5.99

10 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA and class obtained.

Appendix – I

Details of BBA – Business Analytics Programme Structure, Credit Requirements and Specialisation

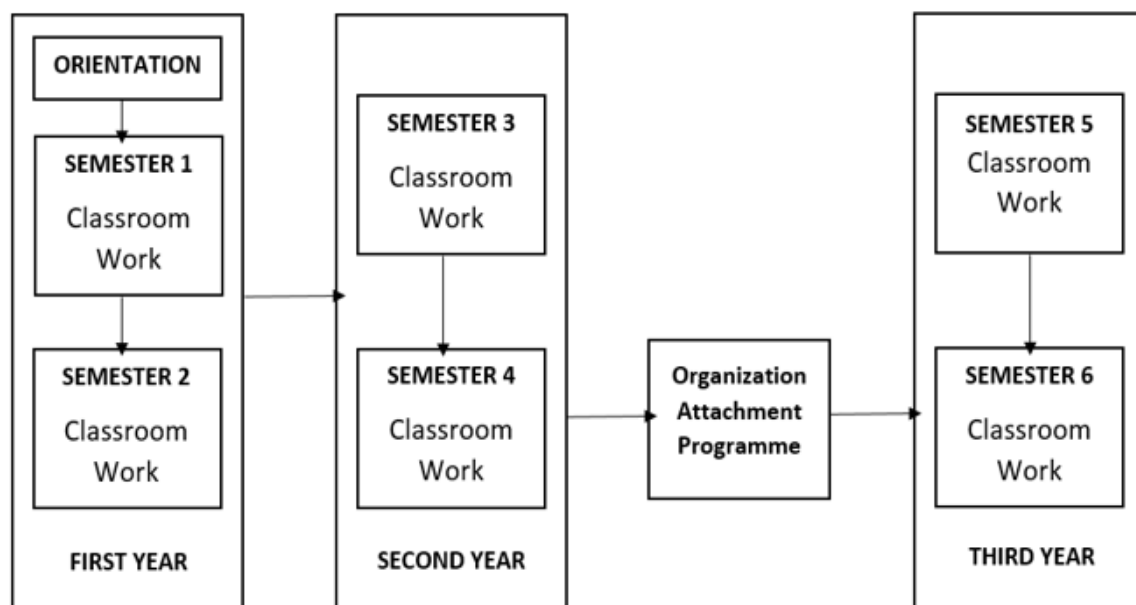
A1. Programme Structure

A1.1 The programme is structured into *six semesters*, consisting of classroom contact-based work and *Organization Attachment/Internship Programme*.

A1.2 Each semester will be for a minimum of *90 working days* for classroom work, covering classroom contact sessions, laboratory/tutorial/library/group work, case discussions and presentation, field-based as well as library/internet search-based assignments and projects, classroom exercises, management and simulation games, short quizzes, and class tests. The duration for any organizational attachment/training during the semester and final University examinations will be in addition to the 90 working days.

A1.4 The structure of the BBA programme is as shown in the following figure:

Figure A1: BBA – Business Analytics Programme Structure



A2. Credits

A2.1 Any student of the BBA – Business Analytics programme who earns 140 credits by pursuing the prescribed course work and passing all tests, examinations, assignments, laboratory work, projects and all other evaluation components as per the passing standards of the University will be eligible for the award of the Bachelor of Business Administration (BBA).

- A2.2 The current distribution of credits over the Three-year period for classroom contact sessions and laboratory/ tutorial/ library/ group work sessions will be as follows:

TableA2: Semester-wise Distribution of Credits

Sl. No.	Semester	Number of Credits
1	Semester - 1	20
2	Semester - 2	22
3	Semester - 3	24
4	Semester - 4	24
5	Semester - 5 (Organization Attachment Programme) (18+4)	22
6	Semester – 6	20
	Total Credits	132

- A2.3 Some courses will have only classroom contact sessions and some others will have tutorial/ laboratory/ library/ group work sessions, as shown in the list of courses.

A3. *Courses, Curricula and Revision*

- A3.1 The Faculty Board of the Faculty of Management Studies and the Dean of the Faculty of Management Studies will keep the curricula current and in tune with the changes happening in the world of management and make it relevant to the needs of different organs of society.
- A3.2 The review of the programme, its structure, the course curricula, pedagogy and evaluation will be undertaken by the Boards of Studies.

Bachelor of Business Administration – Business Analytics

Sem.	Course Code	Name of the Course	Nature	Credits	Remarks
1	BMB101	Principles and Practices of Management	Theory	04	-
	BMB102	Logical Analysis and Decision Making	Theory	04	-
	BMB103	Computer Fundamentals and Applications	Practical	04	-
	BMB104	Skills for Management Professionals	Practical	04	-
	HS101.02 D	Communicative English	Practical	02	Offered by FOH
	CL144.01 D	Environmental Sciences	Practical	02	Offered by FTE
Total Credit for the Semester:				20	
2	BMB201	Fundamentals of Economics	Theory	04	-
	BMB202	Introduction to Data Science	Theory	04	-
	BMB203	Basics of Quantitative Methods	Theory	04	-
	BMB204	Introduction to Computing	Practical	04	-
	BMB205	Business Case Analysis	Theory	04	-
	HS201.02 D – HS210.02 D	A Course on Liberal Arts	Practical	02	Offered by FOH
Total Credit for the Semester:				22	
Total Credits for First Year				42	
3	BMB301	Financial Management	Theory	04	-
	BMB302	Human Resources Management	Theory	04	-
	BMB303	Operations Research Models	Theory	04	-
	BMB304	Database Management Systems	Practical	04	-
	BMB305	Marketing Management	Theory	04	-
	HS121.02 D	Creativity, Problem Solving & Innovations	Practical	02	Offered by FOH
	BM231	Banking & Insurance (UE)	Practical	02	University Elective
Total Credit for the Semester:				24	

Sem.	Course Code	Name of the Course	Nature	Credits	Remarks
4	BMB401	Fundamentals of Research	Theory	04	-
	BMB402	Advanced Quantitative Methods	Practical	04	-
	BMB403	Entrepreneurship Development and Innovation Management	Practical	04	-
	BMB404	Predictive Business Analytics using R	Practical	04	
	BMB405	Data Visualization and Communication	Practical	04	
	HSIII.02 D	Human Values & Professional Ethics	Practical	02	<i>Offered by FOH</i>
	BM241	Healthcare Management (UE)	Practical	02	<i>University Elective</i>
Total Credit for the Semester:				24	
Total Credits for Second Year				48	
5	BMB501	Organizational Attachment Program	Practical	04	-
	BMB502	Strategic Management	Theory	04	-
	BMB503	Supply Chain & Logistic Management	Theory	04	-
	BMB504	Machine Learning	Practical	04	-
	BMB505	Analytics for Marketing	Practical	04	-
	HS131.02 D	Communication and Soft Skills	Practical	02	<i>Offered by FOH</i>
Total Credit for the Semester:				22	
6	BMB601	Social Media & Web Analytics	Practical	04	-
	BMB602	Grand Project	Practical	06	-
	BMB603	Business Forecasting for Time-Series	Practical	04	-
	BMB604	Analytics for HR	Practical	04	-
	HS132.02 D	Contributory Personality Development	Practical	02	<i>Offered by FOH</i>
Total Credit for the Semester:				20	
Total Credits for Third Year				42	
Total Credits of 3 Years' BBA Degree Programme				132	

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-I

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BMB101	Principles and Practices of Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB102	Logical Analysis and Decision Making	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB103	Computer Fundamentals and Applications	04	---	4.00	---	4.00	50	50	100	---	---	---
BMB104	Skills for Management Professionals	04	---	4.00	---	4.00	50	50	100	---	---	---
HS101.02 D	Communicative English	02	---	2.00	---	2.00	---	---	---	25	25	50
CL144.01 D	Environmental Sciences	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	20			---	20.00			400			100

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-II

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BMB201	Fundamentals of Economics	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB202	Introduction to Data Science	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB203	Basics of Quantitative Methods	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB204	Introduction to Computing	04	---	4.00	---	4.00	50	50	100	---	---	---
BMB205	Business Case Analysis	04	4.00	---	---	4.00	50	50	100	---	---	---
HS201.02 D – HS210.02 D	A Course on Liberal Arts	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	22				22.00			500			50

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-III

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BMB301	Financial Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB302	Human Resources Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB303	Operations Research Models	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB304	Database Management Systems	04	---	4.00	---	4.00	50	50	100	---	---	---
BMB305	Marketing Management	04	4.00	---	---	4.00	50	50	100	---	---	---
HS121.02 D	Creativity, Problem Solving & Innovations	02	---	2.00	---	2.00	---	---	---	25	25	50
BM231	Banking & Insurance (UE)	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	24				24.00			500			100

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-IV

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BMB401	Fundamentals of Research	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB402	Advanced Quantitative Methods	04	---	4.00	---	4.00	50	50	100	---	---	---
BMB403	Entrepreneurship Development and Innovation Management	04	---	4.00	---	4.00	50	50	100	---	---	---
BMB404	Predictive Business Analytics using R	04	---	4.00	---	4.00	50	50	100	---	---	---
BMB405	Data Visualization and Communication	04	---	4.00	---	4.00	50	50	100	---	---	---
HSIII.02 D	Human Values & Professional Ethics	04	---	4.00	---	4.00	50	50	100	---	---	---
BM241	Healthcare Management (UE)	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	22				22.00			600			50

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-V

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BMB501	Organizational Attachment Program	04	---	4.00	---	4.00	---	---	---	50	50	100
BMB502	Strategic Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB503	Supply Chain & Logistic Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BMB504	Machine Learning	04	---	4.00	---	4.00	---	---	---	50	50	100
BMB505	Analytics for Marketing	02	---	2.00	---	2.00	---	---	---	25	25	50
HS131.02 D	Communication and Soft Skills	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	20				20.00			200			300

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-VI

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BMB601	Social Media & Web Analytics	04	---	4.00	---	4.00	---	---	---	50	50	100
BMB602	Grand Project	06	---	6.00	---	4.00	---	---	---	50	50	100
BMB603	Business Forecasting for Time-Series	04	---	4.00	---	4.00	---	---	---	50	50	100
BMB604	Analytics for HR	04	---	4.00	---	4.00	---	---	---	50	50	100
HS132.02 D	Contributory Personality Development	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	20				20.00						450

Bachelor of Business Administration (BBA) - General Programme

SYLLABI (Semester – III)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

BMB301: FINANCIAL MANAGEMENT
YEAR 2, SEMESTER – III

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

Financial management provides pathways to attain goals and objectives in an organization. The key role of a financial manager is to measure organizational efficiency through proper allocation, acquisition and management. It assists in acquiring funds from different sources. Helps in investing an appropriate amount of funds. Financial management assists in increasing organizational efficiency. Enabling the management in cutting down financial costs by reducing cost of funds, it helps in achieving financial goals of an organization. CFO/Finance Manager always ensures optimum utilization of funds. It helps business firm to take financial decisions and preparing guidelines for earning maximum profits with minimum cost. Management gets significant information through financial reporting

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Financial Management	07
2	Business Finance	09
3	Cost of Capital	11
4	Leverage Analysis	11
5	Financial Statement Analysis	10
6	Investment Appraisal	12
Total Contact Hours		60

Course Syllabus

Module 1: Introduction to Financial Management

07 hours

Forms of business Organization, Financial decisions, Scope of financial Management, Profit maximization objective vs. wealth maximization objective

Module 2: Business Finance

09 hours

Short term Sources of Raising Finance, Long term Sources of Raising Finance, Modern Sources of Funds

Module 3: Cost of Capital

11 hours

Meaning and significance of cost of capital, Explicit cost and Implicit cost, Cost of Debt, Cost of Preference Shares and Cost of Equity, Weighted Average cost of capital (WACC)

Module 4: Leverage Analysis

11 hours

Meaning and concept of Leverage, Benefits and Drawbacks of Leverage, Computation and inferences of Degree of Operating Leverage, Financial Leverage and Combined Leverage

Module 5: Financial Statement Analysis

10 hours

Comparative Statements, Common Size Statements, Trend Analysis

Module 6: Investment Appraisal

12 hours

Evaluating Investment Proposals for decision Making, Non-Discounted cash flow methods, Discounted cash flow methods

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 3 hours based on written paper for 70 marks.

Text Book:

1. Prasanna Chandra, Fundamentals of Financial Management (Latest edition), Tata McGraw Hill

Reference Books:

1. Khan and Jain, Financial Management (Latest edition), Tata McGraw Hill
2. I.M. Pandey, Financial Management (Latest edition), Vikas Publishing House
3. Shashi K. Gupta, R.K. Sharma, Financial Management Theory and Practice (Latest edition), Kalyani Publishers

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understanding important financial decisions concerning; financing, investing, dividend, and liquidity.
CO2	Potential to make decisions with regard to maximizing wealth.
CO3	Assists organizations in planning, acquisition and optimum utilization of funds.
CO4	Analyzing and interpreting financial statements to provide an optimum business solution.
CO5	Effective investment appraisal through several tools and techniques enabling the organization to achieve its long-term goals.

Course Outcomes Mapping

Module	Module Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Financial Management	√				
2	Business Finance	√		√		
3	Cost of Capital		√	√		
4	Leverage Analysis		√			
5	Financial Statement Analysis				√	
6	Investment Appraisal					√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understanding</i> important financial decisions concerning; financing, investing, dividend, and liquidity.	2	2	2	3
CO2	<i>Potential</i> to make decisions with regard to maximizing wealth.	3	3	2	3
CO3	<i>Analyzing</i> and interpreting financial statements to provide an optimum business solution.	3	3	2	3
CO4	<i>Assists</i> organizations in planning, acquisition and optimum utilization of funds.	3	3	2	3
CO5	<i>Effective</i> investment appraisal through several tools and techniques enabling the organization to achieve its long-term goals.	3	3	2	3

BMB302: HUMAN RESOURCES MANAGEMENT
YEAR 2, SEMESTER – III

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

Activities of Human Resource Management begins prior to the entry of individuals in an organization and continues till the person leaves the organization. This course will enable students to understand the basic concepts and processes of human resource management in any organization. Students will acquire knowledge of basic HRM functions such as recruitment, selection, performance appraisal, reward and recognition etc

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Need and Concept of HRM	10
2	Human Resource Planning	10
3	Recruitment	10
4	Employee Training and Development & Performance Appraisal	10
5	Compensation Planning	10
6	Industrial Relations	10
Total Contact Hours		60

Course Syllabus

Module 1: Need and Concept of HRM **10 hours**

Concept, Philosophy, and Evolution of HRM, Functions and Scope of HRM, Human Resource Policies, Importance and Objectives

Module 2: Human Resource Planning **10 hours**

Concept, Need, Responsibility for HR Planning, Assessing Human Requirements, Human Resource Forecasting, Job Analysis, Job Description and Job Specification.

Module 3: Recruitment **10 hours**

Meaning, Pre-Requisites of a Good Recruitment Policy, Various Sources of Recruitment, Steps in Selection Procedure, Psychological Testing, Purpose, Characteristics and Various Types of Tests, Interviews – Types, Induction, Objectives and Procedure

Module 4: Employee Training and Development & Performance Appraisal **10 hours**

Need Importance and Objectives, Steps in Training Programme, Training Policy, Different Training Techniques, Executives/Management Development Process, Methods of Performance Appraisal, Possible Errors in Appraisal Process, Planning for Performance Improvement

Module 5: Compensation Planning **10 hours**

Objective of Compensation Planning, Theories underlying Motivation and Remuneration Developing Pay Structures, Principles of Wage and Salary Administration

Module 6: Industrial Relations **10 hours**

Workers Participation in Management, Theories of Industrial Relations, HRM Practices - Change in Perspective, Collective Bargaining - Policies and Practices

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 3 hours based on written paper for 70 marks.

Text Book:

1. K. Aswathappa, Human Resource Management, Latest Edition Himalaya Publishing House.

Reference Books:

1. Garry Dessler, Human resource management, Latest Edition, PHI
2. C B Gupta, Human resource management, Latest Edition), Sultan Chand and Sons
3. Muller Camen Croucher Leigh, Human Resource Management, Latest Edition), Jaico Publishing House

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Develop understanding of basic concepts of human resource management
CO2	Demonstrate understanding of various functions of HRM and the linkages between same
CO3	Analyze the impact of decision making in HRM functions
CO4	Develop Mind-set for HR as a value added function.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Need and Concept of HRM	√			
2	Human Resource Planning		√		
3	Recruitment			√	
4	Employee Training and Development & performance Appraisal				√
5	Compensation Planning				
6	Industrial Relations				

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution	<i>Practice</i> effective communication and interpersonal skills	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity	<i>Demonstrate</i> Attitude for continuous learning and development
CO1	<i>Develop</i> understanding of basic concepts of human resource management	1	1	3	2
CO2	<i>Demonstrate</i> understanding of various functions of HRM and the linkages between same	2	1	3	2
CO3	<i>Analyze</i> the impact of decision making in HRM functions	2	2	3	3
CO4	<i>Develop</i> Mind-set for HR as a value added function.	1	1	3	1

BMB303: OPERATIONS RESEARCH MODELS YEAR 2, SEMESTER – III

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

The course will explore the approaches of operations management and how it can be useful for making strategic decisions. It helps the student to formulate the business related problems in terms of mathematical models and get plausible solution to it.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Operations Research	10
2	Project Analysis, Selection and Scheduling	10
3	Theory of Games	10
4	Linear Programming	10
5	Transportation and Assignment Problem	10
6	Queuing System & Decision Tree	10
Total Contact Hours		60

Course Syllabus

Module 1: Introduction to Operations Research **10 hours**

Meaning and scope of Operations Research 1.2 Applications of OR in Business, Limitations of Operations Research, Replacement Theory: Constant and changes with time.

Module 2: Project Analysis, Selection and Scheduling **10 hours**

Project Initiation and resource allocation, Market analysis, Demand analysis, Technical Analysis, Project Network, Representation, Network Diagram, Project Plan

Module 3: Theory of Games **10 hours**

Basic Concept and Definitions, Terminologies in Game Theory, The theory of rational choice, Two Person Zero Sum Game

Module 4: Linear Programming **10 hours**

Definition, Characteristics of LP, Problem formulation, Graphical solution, Simplex method

Module 5: Transportation and Assignment Problem **10 hours**

Transportation Problem: Formulation, Optimal Solution, Balanced Transportation problem, Assignment problem, formulation, optimal solution, variants of assignment

Module 6: Queuing System & Decision Tree **10 hours**

Essential Features of a Queuing System, Performance Measures of a Queuing System, Classifications of Queuing Models and Their Solutions Introduction to decision tree

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 3 hours based on written paper for 70 marks.

Text Book:

1. Projects- Planning, Analysis, Implementation and Control, Prasanna Chandra, Tata Mc Graw Hill, New Delhi. ISBN: 0074620495

Reference Books:

1. Operations Research - Theory and Applications, J.K Sharma, Macmillan Publications India Ltd, ISBN: 9789350593363
2. Operations Research, H.A.Taha, Pearson ISBN : 0134444019
3. Game Theory: A Very Short Introduction, Ken Binmore, Oxford, ISBN: 0199218463
4. Operations Research, S. Kalavathy, Vikas Publishing House, ISBN: 9325963477

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the meaning, purpose, and tools of Operations Research.
CO2	Apply the applications of Operations Research.
CO3	Create optimal solutions to industry specific problems.
CO4	Implement solutions to business problems using OR methods.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Introduction to Operations Research	√			
2	Project Analysis, Selection and Scheduling		√		
3	Theory of Games		√		
4	Linear Programming			√	
5	Transportation and Assignment Problem			√	
6	Queuing System & Decision Tree				√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross-Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understand</i> the meaning, purpose, and tools of Operations Research.	3	3	-	-
CO2	<i>Apply</i> the applications of Operations Research.	-	1	3	3
CO3	<i>Create</i> optimal solutions to industry specific problems.	-	2	3	3
CO4	<i>Implement</i> solutions to business problems using OR methods.	-	2	3	3

BMB304: DATABASE MANAGEMENT SYSTEM YEAR 2, SEMESTER – III

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	04	60

Rationale

In today's scenario, Information Technology (IT) is essential for each and every individual unit of any businesses. These technologies are used to develop through Database which requires data and its datasets to integrate, design and collaborate the various functional areas of business through Database management information system. Therefore, this curriculum course structure enhances the student is able to understand, identify, design, deploy and implement an effective Database Management system through the use of various Software's and its components.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Fundamentals of Database	10
2	Databases Relationships	10
3	Database Relational Model	10
4	Structured Query Language	10
5	Database Security	10
6	Introduction of MS Access	10
Total Contact Hours		60

Course Syllabus

Module 1: Fundamentals of Database

10 hours

Database Management System: Components of data processing system, Three level architecture of data base management system (Data Abstraction), Relational data model, Network data model, Hierarchical model.

Module 2: Databases Relationships

10 hours

Conceptual Data Modelling using Entities and Relationships, Entity types, Entity sets, attributes, roles, and structural constraints, Weak entity types, ER diagrams, examples, Specialization and Generalization

Module 3: Database Relational Model

10 hours

Introduction, Terms: Relation, Tuple, Attribute, Cardinality, Degree of relationship set, Domain, Keys: Super Key, Candidate Key, Primary Key, Foreign Key, Relational Algebra Operations: Select, Project, Union, Difference, Intersection, Cartesian Product, Natural Join

Module 4: Structured Query Language

10 hours

SQL: SQL data definition and data types, specifying constraints in SQL, Retrieval queries in SQL, INSERT, DELETE, and UPDATE statements in SQL, Additional features of SQL and queries

Module 5: Database Security

10 hours

Authorization is SQL (privileges in SQL), Roles, The privilege to Grant Privileges, Limitations of SQL authorization

Module 6: Introduction of MS Access

10 hours

MS Access – Overview, MS Access, Data Types, Create Database, Create Tables, Adding Data, Query Data, Query Criteria, Action Queries, Create Relationships, One-To-One, One-To Many, Many-To-Many, Built-In Functions.

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software based analysis depending on the nature of the course.

Text Book:

1. Database System Concepts, Silberschatz, Korth, McGraw hill, Sixth Edition, ISBN: 9332901384.

Reference Books:

1. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, TATA McGrawHill, 3rd Edition, ISBN: 9780071231510

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the basic concepts and the applications of database systems
CO2	Demonstrate the working of different concepts of DBMS and establish relation among the datasets.
CO3	Practice Structured Query Language (SQL) for database Creation and manipulation.
CO4	Construct a database by using data definition, data manipulation and control languages.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Fundamentals of Database	√			
2	Databases Relationships		√		
3	Database Relational Model			√	
4	Structured Query Language			√	
5	Database Security	√			
6	Introduction of MS Access				√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		Analyze contemporary issues and apply management knowledge for resolution.	Practice effective communicatio n and interpersonal skills.	Understand Global Perspectives and exhibit Cross- Cultural sensitivity.	Demonstrate Attitude for continuous learning and development.
CO1	Understand the basic concepts and the applications of database systems	2	1	1	1
CO2	Demonstrate the working of different concepts of DBMS and establish relation among the datasets.	2	2	3	3
CO3	Practice Structured Query Language (SQL) for database Creation and manipulation.	3	3	1	1
CO4	Construct a database by using data definition, data manipulation and control languages.	3	3	3	3

BMB305: MARKETING MANAGEMENT
YEAR 2, SEMESTER – III

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

Marketing is the key to establish any business successfully. Marketing managers need to develop the skills of understanding their customers by creating awareness to their offerings and persuading them to buy it. Additionally, marketing managers have to deal with complex procedure of all strategic marketing activities which is not limited to persuading the stakeholders rather convincing them for establishing sustainable long-term relationship. This curriculum course structure is developed such that the student is able to understand the basic understanding of marketing and marketing environment to create ultimate value to their customers in the competitive environment.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Conceptual Foundation & Marketing Environment	08
2	Analyzing Consumers and Selecting Markets	11
3	Creating Customer Value: Managing the Product	12
4	Delivering Customer Value: Managing Distribution	12
5	Communicating Customer Value: Integrated Marketing Communication	12
6	Capturing Value & Supporting Marketing Efforts	05
Total Contact Hours		60

Course Syllabus

Module 1: Conceptual Foundation & Marketing Environmental **08 hours**

Understanding Marketing as a Value-delivering Task, Marketing Environment, Contemporary Marketing Environment, Marketing Strategies.

Module 2: Analyzing Consumers and Selecting Markets **11 hours**

Understanding the Consumers, Consumer Behavior, Buying Decision Process, Segmentation and Targeting for Maximizing.

Module 3: Creating Customer Value: Managing the Product **12 hours**

Product Management, Managing Services, Managing Brands and Brand Equity.

Module 4: Delivering Customer Value: Managing Distribution **12 hours**

Managing Distribution Logistics, Designing and Managing Marketing Channel, Direct Marketing & Digital Marketing.

Module 5: Communicating Customer Value: Integrated Marketing Communication **12 hours**

Managing Mass Communication, Advertising, Sales Promotion, Digital Communication, Managing Personal Communication, Concept of Personal Selling, Sales Management, Customer Relationship Management.

Module 6: Capturing Value & Supporting Marketing Efforts **05 hours**

Pricing, MIS and Marketing Research, Special Fields in Rural Marketing.

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 3 hours based on written paper for 70 marks.

Text Book:

1. Philip Kotler, Marketing Management, Pearson Publication Latest Edition, South Asian Perspective.

Reference Books:

1. Philip Kotler, Marketing Management, Routledge, Latest Edition, Prentice Hall of India Publications.
2. Kumar Arun and Meenakshi, Marketing Management, Latest Edition, Vikas Publishing.
3. Dalrymple, J.D. and Parson, J.L.; John Wiley and Sons, Marketing Management Strategy and Cases, Latest Edition, Tata McGraw Hill.

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand and Use of the fundamentals of marketing concept and its role in business.
CO2	Evaluate the marketing concepts in relation to consumers' market in strategic value creation.
CO3	Evaluate the existing marketing channel design and digital and direct marketing strategies.
CO4	Choose the appropriate marketing communication and information system and strategy for creating utmost customer value.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Conceptual Foundation & Marketing Environment	√			
2	Analyzing Consumers and Selecting Markets		√		
3	Creating Customer Value: Managing the Product			√	
4	Delivering Customer Value: Managing Distribution				√
5	Communicating Customer Value: Integrated Marketing Communication			√	
6	Capturing Value & Supporting Marketing Efforts				√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understand</i> and Use of the fundamentals of marketing concept and its role in business.	1	1	1	1
CO2	<i>Evaluate</i> the marketing concepts in relation to consumers' market in strategic value creation.	1	2	1	1
CO3	<i>Evaluate</i> the existing marketing channel design and digital and direct marketing strategies.	1	1	1	1
CO4	<i>Choose</i> the appropriate marketing communication and information system and strategy for creating utmost customer value.	1	1	1	1

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
FACULTY OF HUMANITIES (FOH)
Semester- III

HS121.02 D: CREATIVITY, PROBLEM SOLVING AND INNOVATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	02	2
Marks	--	100	--	100	

Pre-requisite courses:

Creative Problem Solving

<https://www.coursera.org/learn/creative-problem-solving>

Objectives of the Course:

To facilitate learners to:

- gain familiarity with the mechanics of creativity and problem solving.
- develop an attitude for innovation.
- develop creative thinking skills using cone of learning components leading to understanding of strategies of creativity, problem solving and innovation.
- explore applications of the concepts of creativity and problem-solving skills in personal, social, academic, and profession life.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1.	Introduction to Creativity, Problem Solving and Innovation	06 Hours	20%
	Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills		
2.	Questioning, Learning and Visualization	6 Hours	20%
	Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education		

	in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking		
3.	Creative Thinking and Problem Solving	6 Hours	20%
	Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving		
4.	Logic, Language and Reasoning	6 Hours	20%
	Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning		
5.	Contemporary Issues and Practices in Creativity and Problem Solving	6 Hours	20%
	Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving		

Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Puzzles, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

Evaluation:

There will be no end semester university examinations. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students distribute the marks (out of 30 as Internal and out of 70 as External) and submit them.

Evaluation Scheme

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Attendance	100 %	--	20
2	Individual Activity Participation	As stipulated by the Resource Person(s) in the Training		20
3	Group Activity Participation			20
4	Presentation			30
5	Feedback on Improvement			10
Total				100

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Recommended Study Material:**❖ Text book:**

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

❖ Reference book:

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map, Thorsons Publication 2002
5. Scott Berkun, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992
8. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

❖ Additional Readings

1. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
5. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994

9. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
11. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds, Basic Books Publication 1993
14. Mihaly Csikszentmihalyi, Creativity–Flow and Psychology of Discovery and Invention, Harper Publication 1996
15. Martin Gerdner, W. H., Aha! Insight, Freeman Publication 1978
16. Paul Slovic, Test Your Lateral Thinking IQ, Sterling Publication 1994
17. Paul Slovic & Des Macale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

❖ **Web material:**

1. Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

BM231: BANKING AND INSURANCE
YEAR 2, SEMESTER – III

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)	Course Classification
02	Practical	02	30	University Elective

Rationale

Banking and Insurance sector witness a robust growth in the financial service market. Due to advent of digital technologies coupled with favourable regulatory environment, financial institutions have developed capabilities to offer wide variety of products and services to the customers. This course will enable the students to understand the different banking and insurance products and services along with brief overview of banking and insurance sector outlook in India.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Indian Banking Sector	05
2	Bank Management	05
3	Overview of Retail Banking	05
4	Payment and Settlement System	05
5	Introduction to Insurance Sector	05
6	Marketing of Banking and Insurance Services	05
Total Contact Hours		30

Course Syllabus

Module 1: Introduction to Indian Banking Sector **05 hours**

Evolution of Banking, Reserve Bank of India and its roles in Indian financial sector, Brief Structure of Banks, Commercial Banking and Universal Banking, Types of Customers

Module 2: Bank Management **05 hours**

Sources and Uses of Funds in Banks, Value Chain Analysis in Banking Industry, Leveraging New Technologies in Banking, Methods of Credit Control & Credit Creation, Financial Statements of Bank

Module 3: Overview of Retail Banking **05 hours**

Retail Banking –Concept and Importance, Retail Banking Products (Liability focused segment), Retail Banking Products (Asset focused segment)

Module 4: Payment and Settlement System **05 hours**

Payment & Settlement System – New Age Clearing (SI, ECS), Payment & Settlement System – National Gateways (RTGS, NEFT), Digital Financial Services: Unified Payments Interface (UPI), Digital Wallets (E-Wallets), IMPS etc.

Module 5: Introduction to Insurance Sector **05 hours**

Concept of Insurance: Relevance of Insurance, IRDAI and Its role, Principles of Insurance, Types of Insurance services (Life Insurance and General Insurance), Different product offerings under life and general insurance and its tax implications.

Module 6: Marketing of Banking and Insurance Services **05 hours**

Marketing Strategy of Banking and Insurance Services, SERVQUAL Model, Role of technologies in Banking and insurance service offerings

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

- The Internal Evaluation Component shall be shared by the course coordinator.

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software based analysis depending on the nature of the course.

Text Book:

1. Indian Institute of Banking and Finance, Principles and Practices of Banking (Latest Edition), Macmillan Education, ISBN: 978-9387687240.

Reference Books:

1. Indian Institute of Banking and Finance, Retail Banking, (Latest Edition), Macmillan Education, ISBN: 978-9388175838
2. Indian Institute of Banking and Finance, Bank Financial Management, (Latest Edition), Macmillan Education, ISBN: 978-9387000650

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the basic concept of banking and insurance as a financial intermediation service and bank/insurance companies as a financial institution.
CO2	Explore various functions and services offered by banking and insurance companies
CO3	Analyze the financial statements of banks for decision making
CO4	Explore the role of technology in banking and insurance product offerings and marketing.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Introduction to Indian Banking Sector	√			
2	Bank Management		√	√	
3	Overview of Retail Banking		√		
4	Payment and Settlement System				√
5	Introduction to Insurance Sector	√			
6	Marketing of Banking and Insurance Services	√			

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communicatio n and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understand the basic concept of banking and insurance as a financial intermediation service and bank/insurance companies as a financial institution.</i>	3	1	2	3
CO2	<i>Explore various functions and services offered by banking and insurance companies.</i>	3	2	3	3
CO3	<i>Analyze the financial statements of banks for decision making.</i>	3	3	2	3
CO4	<i>Explore the role of technology in banking and insurance product offerings and marketing.</i>	3	1	3	3

Bachelor of Business Administration (BBA) - General Programme

SYLLABI (Semester – IV)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

BMB401: FUNDAMENTALS OF RESEARCH YEAR 2, SEMESTER – IV

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

Managers have to deal with important decision making. Decision making is a complex process of choosing the alternative solution to resolve any problem and in this problem solving, research play important role to help and generate viable alternatives for effective decision making. This curriculum course structure of ‘Fundamentals of Research’ is developed such that the student would be able to evaluate, identify the critical issues, gathering relevant information, analysing the data that would help in effective decision making and implementing the right course of action.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students’ prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Research	09
2	Defining the Research Problem	11
3	Research Design	12
4	Sampling and Questionnaire Design and Construction	11
5	Measurement and Scaling, Data Collection and Preparation	12
6	Data Analysis with SPSS and Application of Research, Report Writing	05
Total Contact Hours		60

Course Syllabus

Module 1: Introduction to Research

09 hours

Meaning and Definition of Research, Objectives and Significance of Conducting Research, Types of Research, Typical Research Process, General Problems encountered by Researcher while conducting research, Criteria of Good Research, Literature Review and its Importance.

Module 2: Defining the Research Problem

11 hours

Understanding The Meaning of Research Problem, Research Field, Research Area and Research Topic, Identifying and selecting the research problem, Techniques Involved in Defining a Problem.

Module 3: Research Design

12 hours

Meaning and need of research design, Good Research design characteristics features, Important concepts related to research design, Types of research and related research design, Special Case of Experimental Research Design, Basic Principles and types of Experimental design.

Module 4: Sampling Design

11 hours

Introduction to sampling and sampling design, Sampling framework and important considerations, Types of sampling design, Sampling and Non sampling error.

Module 5: Measurement and Scaling, Data Collection and Preparation

12 hours

Qualitative and Quantitative Data and Measurement Scale, Classifications of Measurement Scale and its Bases, Techniques of Developing Measurement Scale, Goodness of Measurement Scale and Possible Sources of Measurement Scale, Questionnaire Designing, And different Approaches of data collection, Qualitative and Quantitative method of data collection Data Preparation (Editing, Coding, Classification, Tabulation)

Module 6: Data Analysis with SPSS and Application of Research, Report Writing

05 hours

Descriptive data analysis, Hypothesis testing Correlation, Chi-Square test, Analysis of Variance, Linear Regression analysis, Significance of Report Writing, Types of report writing, Steps in report writing

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
4otal marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 3 hours based on written paper for 70 marks.

Text Book:

1. Kothari and Gaurav Garg, Research Methodology Methods and Technique, New Age International (P) Ltd, ISBN-13 978-9386649225

Reference Books:

1. Bryman Alan, Bell Emma, Business Research Methods, Oxford University Press, ISBN: 9780199668649.
2. Zikmund, Babin, Carr, Adhikari, Griffin, Business Research Methods, A South Asian Perspective, Cengage Learning, ISBN: 9781111826925

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the Use of the research and literature review in understanding the problem situation
CO2	Interpret the information available in the problem situation
CO3	Evaluate companies' failure due to lack of research, thus the importance of research and other ways of obtaining the answers to the problems.
CO4	Evaluate the research design and appropriateness of the sampling design in a given context.
CO5	Interpret and choose the findings as a best solution to the identified problem.

Course Outcomes Mapping

Module	Module Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Research	√				
2	Defining the Research Problem		√			
3	Research Design			√		
4	Sampling and Questionnaire Design and Construction				√	
5	Measurement and Scaling, Data Collection and Preparation					√
6	Data Analysis with SPSS and Application of Research, Report Writing					√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		Analyze contemporary issues and apply management knowledge for resolution.	Practice effective communication and interpersonal skills.	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.	Demonstrate Attitude for continuous learning and development.
CO1	Understand the Use of the research and literature review in understanding the problem situation.	2	2	2	2
CO2	Interpret the information available in the problem situation	3	2	2	1
CO3	Evaluate companies' failure due to lack of research, thus the importance of research and other ways of obtaining the answers to the problems.	1	1	1	1
CO4	Evaluate the research design and appropriateness of the sampling design in a given context.	3	2	2	2
CO5	Interpret and choose the findings as a best solution to	1	1	1	1

BMB402: ADVANCED QUANTITATIVE METHODS
YEAR 2, SEMESTER – IV

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	04	60

Rationale

Advanced analytics has evolved into a competitive strategy that drives innovation across several organizations. This course will introduce the students to advanced statistical methods and expose them to applications of the same for decision making.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Correlation and Regression	10
2	Logistic Regression	10
3	Decision Trees	10
4	Forecasting Techniques	10
5	Clustering	10
6	Contemporary Issues	10
Total Contact Hours		60

Course Syllabus

Module 1: Correlation and Regression

09 hours

Introduction to correlation, Pearson correlation coefficient, Spurious correlation, Introduction to regression, Simple linear regression, Multiple linear regression

Module 2: Logistic Regression

11 hours

Introduction to logistic regression, Binary logistic regression, Interpretation of logistic regression parameters, Classification table, sensitivity, specificity, precision, and F-Score
Optimal cut-off probability, Introduction to multinomial logistic regression

Module 3: Decision Trees

12 hours

Introduction, Chi-square automatic interaction detection (CHAID), Classification and regression tree, Error matrix and AUC

Module 4: Forecasting Techniques

11 hours

Introduction, Time-series data and components of time-series data, Forecasting techniques and forecasting accuracy

Module 5: Clustering

12 hours

Introduction, Distance and similarity measures, Quality and optimal number of clusters, Clustering algorithms

Module 6: Contemporary Issues

05 hours

Contemporary Issues in advanced statistical analytics

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software based analysis depending on the nature of the course.

Text Book:

1. Business Analytics, U Dineshkumar, (Latest Edition), Wiley, ISBN: 9789354246197

Reference Books:

1. Applied Statistics in Business and Economics, Donae Seward and Chowdhury, (Latest Edition), McGraw Hill, ISBN: 9789390113057

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Learn the rationale behind the advanced statistical methods and its importance in decision making
CO2	Apply the appropriate advance level statistical techniques for problem solving
CO3	Analyze the data to make inference about the population characteristics.
CO4	Propose models that can be used for the purpose of prediction / forecasting.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Correlation and Regression	√			
2	Logistic Regression		√		
3	Decision Trees			√	
4	Forecasting Techniques				√
5	Clustering				√
6	Contemporary Issues				√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross-Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Learn the rationale behind the advanced statistical methods and its importance in decision making</i>	2	3	2	3
CO2	<i>Apply the appropriate advance level statistical techniques for problem solving</i>	3	3	3	3
CO3	<i>Analyze the data to make inference about the population characteristics.</i>	2	3	3	3
CO4	<i>Propose models that can be used for the purpose of prediction / forecasting.</i>	2	3	3	3

**BMB403: ENTREPRENEURSHIP DEVELOPMENT AND INNOVATION
MANAGEMENT
YEAR 2, SEMESTER – IV**

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	04	60

Rationale

The goals of this Programme are to inspire students and help them imbibe an entrepreneurial mindset. The students will learn what entrepreneurship is and how it has impacted the world and their country. They will be introduced to key traits and the DNA of an entrepreneur, and be given an opportunity to assess their own strengths and identify gaps that need to be addressed to become a successful entrepreneur.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Entrepreneurship	10
2	Theories of Entrepreneurship and Entrepreneurial Motivation	10
3	Opportunity Identification and Developing Successful Business Idea	10
4	Environmental Analysis and Developing and Entrepreneurial Firm	10
5	Financing the Venture	10
6	Institutional Support to Entrepreneurship	10
Total Contact Hours		60

Course Syllabus

Module 1: Introduction to Entrepreneurship

10 hours

Concept of Entrepreneurship, Factors impacting the emergence of entrepreneurship, Entrepreneur Vs Intrapreneur, Common Characteristics attributed to Entrepreneurs, Classification and Types of Entrepreneurs, Entrepreneurial Process, Barriers to Entrepreneurship, Entrepreneurship's Importance, Myths of Entrepreneurship.

Module 2: Theories of Entrepreneurship and Entrepreneurial Motivation

05 hours

Economic, sociological and psychological theories of Entrepreneurship, Entrepreneurship Innovation theories: David McClelland's Theory of Achievement Motivation, Schumpeter's Theory of Innovation and Economic development, Peter Drucker's Theory of Entrepreneurship, Entrepreneurial Motivation- Internal and external factors, Testing Entrepreneurial Motivation and Behaviour.

Module 3: Opportunity Identification and Developing Successful Business Idea

05 hours

Identifying and Recognizing Opportunities, Sources of New Venture Ideas, Idea Generation techniques, Creativity and Entrepreneurship, Entrepreneurial Innovation, Feasibility Analysis.

Module 4: Environmental Analysis Developing and Entrepreneurial Firm

15 hours

Industry Analysis, Competitor Analysis, Business Plan: Concept and Outline of business plan, Business Model: Concept and components of business model, Preparing Ethical and Legal Foundation: Business license, Business permits and forms of Business Organization.

Module 5: Financing the Venture

10 hours

Importance of getting financing or funding, Sources of Equity Funding, Sources of Debt Funding, Creative sources of Financing or Funding.

Module 6: Institutional Support to Entrepreneurship

15 hours

SIDO, SISI, NSIC, SIDCO, DIC – their functions, SIDBI's Schemes, Incentives: Subsidy, Tax Concessions, Marketing and Export Assistance, Sickness – definition, symptoms, causes, Measures to prevent sickness in small units.

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software-based analysis depending on the nature of the course.

Text Book:

1. Bruce R Barringer and R Duanne Ireland, Entrepreneurship: Successful Launching New Ventures, Latest Edition, Pearson Education.

Reference Books:

1. P.C. Jain, Handbook for Entrepreneurs, Entrepreneurship Development of India, Latest Edition, Oxford Publication.
2. Robert D. Hisrich, Michael P. Peters & Dean A. Shepherd, Entrepreneurship, Latest Edition, Tata McGraw Hill Publication.

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Develop awareness about entrepreneurship and successful entrepreneurs.
CO2	Design start-up plans, that work for getting knowledge of how a business starts and take a blended advantage of business analytics for entrepreneurship and intrapreneurship.
CO3	Apply innovative solutions to the problems raised in the business context
CO4	Recognize the funding opportunities for business ventures

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Introduction to Entrepreneurship	√			
2	Theories of Entrepreneurship and Entrepreneurial Motivation		√	√	
3	Opportunity Identification and Developing Successful Business Idea		√	√	
4	Environmental Analysis and Developing and Entrepreneurial Firm		√	√	
5	Financing the Venture				√
6	Institutional Support to Entrepreneurship				√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Develop</i> awareness about entrepreneurship and successful entrepreneurs.	2	1	-	1
CO2	<i>Design</i> start-up plans, that work for getting knowledge of how a business starts and take a blended advantage of business analytics for entrepreneurship and intrapreneurship.	1	2	-	-
CO3	<i>Apply</i> innovative solutions to the problems raised in the business context	1	1	-	-
CO4	<i>Recognize</i> the funding opportunities for business ventures	1	-	-	-

BMB404: PREDICTIVE BUSINESS ANALYTICS USING R
YEAR 2, SEMESTER – IV

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	04	60

Rationale

This course will enable the students gain sufficient knowledge and experience to perform analysis using different analytical tools available in R. This course will help them understand concepts and use the libraries of R for prediction which will help in decision making.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Basics of R	10
2	Visualization Techniques	10
3	Statistical Analysis	10
4	Supervised Machine Learning	10
5	Unsupervised Machine Learning	10
6	Contemporary Issues	10
Total Contact Hours		60

Course Syllabus

Module 1: Basics of R

10 hours

Introduction to R and R-Studio, Data types in R, Programming in R, Data exploration and manipulation, Import and export of data

Module 2: Visualization Techniques

10 hours

Introduction to visualization, Basic visualization, Advanced visualization

Module 3: Statistical Analysis

10 hours

Introduction to basic statistics, Descriptive statistics in R, Compare means in R, Basic time-series models in R

Module 4: Supervised Machine Learning

10 hours

Introduction to supervised machine learning, Linear regression, Naïve-Bayes algorithm, KNN algorithm, SVM algorithm, Random forest

Module 5: Unsupervised Machine Learning

10 hours

Introduction to unsupervised machine learning, Factor analysis, k-Means clustering, Hierarchical clustering

Module 6: Contemporary Issues

10 hours

Contemporary issues in analytics that can be analyzed and solved using R

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software-based analysis depending on the nature of the course.

Text Book:

1. Data Analytics with R, Dr. Bharti Motwani, Latest Edition, Wiley, ISBN: 13 978-8126576463

Reference Books:

1. Data Analytics using R, Seema Acharya, Latest Edition, McGraw Hill, ISBN: 13 978-9352605248

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the foundation of R.
CO2	Use relevant libraries for data analysis.
CO3	Analyze the data to make inference about the population characteristics.
CO4	Propose models that can be used for the purpose of prediction or forecasting.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Basics of R	✓			
2	Visualization Techniques		✓		
3	Statistical Analysis			✓	
4	Supervised Machine Learning				✓
5	Unsupervised Machine Learning				✓
6	Contemporary Issues				✓

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	Understand the foundation of R.	1	1	1	1
CO2	Use relevant libraries for data analysis.	2	2	3	3
CO3	Analyze the data to make inference about the population characteristics.	2	3	3	3
CO4	Propose models that can be used for the purpose of prediction or forecasting.	2	3	3	3

BMB405: DATA VISUALIZATION AND COMMUNICATION
YEAR 1, SEMESTER – IV

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	04	60

Rationale

Modern data visualization technology is causing a paradigm shift in the way organizations convert raw data into actionable information. Visualization facilitates rapid understanding of trends and outliers within datasets. Therefore, this course will provide students with a formal grounding in data visualization as well as hands-on experience using software such as Tableau, PowerBI, etc.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Creating Visual Analytics	10
2	Connecting to Data	10
3	Building first Visualization	10
4	Creating Calculations to Enhance Data	10
5	Forecasting with Tableau	10
6	Tips, Tricks and Time Savers	10
Total Contact Hours		60

Course Syllabus

Module 1: Contemporary Managerial Challenges **10 hours**

Configuring Data Environment, Leveraging Toolbar Icons, Data Window, Data Types & Aggregation, Building Visualization with rows and columns

Module 2: Management of Technological Innovations **10 hours**

Measure Names & Values, Data Extraction, Metrics vs dimensions, Data types and defaults, Aliases and names, Data Visualization Concepts

Module 3: Management of Faith based Organizations **10 hours**

Chart Types, Text Tables, Maps, Trend Lines & Reference Lines, Sorting Data in Tableau Enhancing View with Filters, Set, Groups & Hierarchies

Module 4: Leadership and Diversity in Organizations **10 hours**

Aggregation, Calculated Values & Table Calculations, Building Formula, Calculated fields, Table calculations

Module 5: Non-profit and Volunteer Management **10 hours**

Generating new data with Forecasts, Ad Hoc Analysis with Parameters

Module 6: Research in Administration **10 hours**

Improving Formatting, Improving Appearance, Customizing Shapes & Colors

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

Component	Marks
Quiz	80
Assignment/ Project/ Presentation	90
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	300

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software based analysis depending on the nature of the course.

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Employ best practices in data visualization to develop charts, maps, tables, and other visual representations of data.
CO2	Apply Tableau's visualization features to conduct data analysis, especially exploration of an unfamiliar dataset.
CO3	Create compelling, interactive dashboards to combine several visualizations into a cohesive and functional whole.
CO4	Utilize advanced Tableau features including parameters, data blending, very large datasets, custom date hierarchies, and others.
CO5	Analyze data visualizations, dashboards, and Tableau Stories to support relevant communication for diverse audiences.

Course Outcomes Mapping

Module	Module Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Creating Visual Analytics	√				
2	Connecting to Data		√			
3	Building first Visualization			√		
4	Creating Calculations to Enhance Data				√	
5	Forecasting with Tableau					√
6	Tips, Tricks and Time Savers					√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Employ best practices in data visualization to develop charts, maps, tables, and other visual representations of data.</i>	2	3	2	3
CO2	<i>Apply Tableau's visualization features to conduct data analysis, especially exploration of an unfamiliar dataset.</i>	3	3	3	3
CO3	<i>Create compelling, interactive dashboards to combine several visualizations into a cohesive and functional whole.</i>	2	3	3	3
CO4	<i>Utilize advanced Tableau features including parameters, data blending, very large datasets, custom date hierarchies, and others.</i>	2	3	3	3
CO5	<i>Analyze data visualizations, dashboards, and Tableau Stories to support relevant communication for diverse audiences.</i>	2	3	3	3

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
FACULTY OF HUMANITIES (FoH)
Semester IV

HS111.02 D: HUMAN VALUES AND ETHICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	02	2
Marks	--	100	--	100	

Pre-requisite:

- Basic awareness of Values and Ethics

Objectives of the Course:

To facilitate learners to:

- Develop a familiarity with the Fundamental Human Values and Ethics
- Understanding Values and Ethics as per Ancient Indian Tradition
- Apply Values and Ethics in their professional, social and personal spectrum
- Understand and apply Constitutional Values and Fundamental Duties
- Explore global issues and propose solutions with reference to values and Ethics

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Ethics and Values in Ancient Indian Tradition	08
3.	Types of Values and Applied Ethics	08
4.	Constitutional Values and Fundamental Duties	05
5.	Global Issues of Values and Ethics in Contemporary Society	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1.	Introduction to Values and Ethics	5 Hours	17%
	Need, Relevance and Significance of Values, Concept and Meaning of Values and Ethics		
2.	Ethics and Values in Ancient Indian Tradition	8 Hours	26%
	Lessons from Jatakkatha Lessons from Panchtantra Tales Ethics to be followed from Ancient Hinduism, Jainism and Buddhism Selected Anecdotes from ancient mythology and Moral Learnings		
3.	Types of Values and Applied Ethics	8 Hours	26%
	Personal Values, Social, Civic & Democratic Value, Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics		
4.	Constitutional Values and Fundamental Duties	5 Hours	17%
	Basics of Indian Constitution Constitutional Values: Justice, Liberty, Equality and Fraternity Fundamental Rights Fundamental Duties		
5.	Global Issues of Values and Ethics in Contemporary Society	4 Hours	14%
	Case Studies, Presentations, Projects		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the significance of values and ethics
CO2	Understand concept of values and ethics as per Ancient Indian Tradition
CO3	Understand and apply various types of Values and Applied Ethics
CO4	Develop awareness and initiate implementation of Constitutional Values and Fundamental Duties
CO5	Get aware of the global issues and scope and application of values and ethics in problem solving

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

Reference book:

1. Human Values and Ethics in Workplace, United Nations Settlement Program, 2006.
(http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
2. Ethics for Everyone, Arthur Dorbin, 2009.
(<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>) .
3. Values and Ethics for 21st Century, BBVA.
(https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

❖ Web material:

- www.ethics.org

Note: The subtopics mentioned in the modules 2 and 3 are comprehensive. The faculty member can choose any 2 topics from module and 5 topics from module 3.

BM241: HEALTHCARE MANAGEMENT
YEAR 2, SEMESTER – IV

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)	Course Classification
02	Practical	02	30	University Elective

Rationale

The healthcare business is continuously evolving and growing; therefore, it is essential to stay abreast of the latest developments and manage every part of a hospital or other relevant facility. This course will introduce the students to concepts of management in the health care segment and expose them to several strategies to manage change within the health care industry. Further, it will help the students to understand day-to-day operations of a hospital or department, maintain relationships with stakeholders and handle any issues that arise in the health care sector.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Health care Systems in India	06
2	Hospital Operations Management	06
3	Health Care Economics, finance & Marketing	06
4	Health Care Information System	03
5	Entrepreneurship & Consultancy In Healthcare	06
6	Contemporary Issues	03
Total Contact Hours		30

Course Syllabus

Module 1: Introduction to Health care Systems in India **06 hours**

Health planning in India including various committees and National Health Policy and Health Goals set from time to time, Organized sector with reference to Centre, State, District and Block level structures and local bodies, Primary Health care and concept, Alternative systems of medicine, like Ayurveda, Homeopathy, etc., Holistic Approach Non-Governmental Organizations (NGOs) and Private Voluntary Organizations (PVOs), Unorganized Sector.

Module 2: Hospital Operations Management **06 hours**

Epidemiological basis for healthcare management, Management of Indian Hospitals, Challenges, strategies and modern techniques of hospital management, Health services research & formalized managerial methods.

Module 3: Health Care Economics, finance & Marketing **06 hours**

Health Service Financing and Expenditure Survey, Budgeting, Production and Cost of Health Care, Efficiency Marketing Health and Family Welfare, Marketing for Hospitals, Health Services Marketing; Pharmaceutical Marketing, Patient as a Customer.

Module 4: Health Care Information System **03 hours**

Introduction to HMIS & Its need in a Hospital, Features of HMIS, Categorization of HMIS Modules, Operations Management of HMIS Modules, Advantages of HMIS, Challenges of HMIS, Procurement of HMIS Software, Implementation of HMIS.

Module 5: Entrepreneurship & Consultancy In Healthcare **06 hours**

Overviewing entrepreneurship Resource allocation for the enterprise, Overview of client consultancy, Process of consultancy in healthcare, Feasibility study and action plan, Client counselling and diagnosis system.

Module 6: Contemporary Issues **03 hours**

Contemporary Issues in Health care Management

Course Evaluation

I. Continuous and Comprehensive Evaluation (30 Marks)

- The Internal Evaluation component shall be shared by the course coordinator.

II. Semester End Evaluation (70 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software-based analysis depending on the nature of the course.

Text Book:

1. S C Mohapatra, Meghkanta Mohapatra and Vishvskant Mohapatra, Textbook of Health Management, Innovative Publication (Latest Edition)

Reference Books:

1. Myron Fottler, Donna Malvey and Donna Slovensky, Handbook of Healthcare management, Elgar Publication (Latest Edition),

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Learn the foundation of health care structure and its importance in administrative context.
CO2	Apply the appropriate managerial techniques to resolve the health care challenges.
CO3	Equipping the student to take up entrepreneurship & consultancy work in various facets of healthcare management.
CO4	Analyze the role of information system management in health care segment

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Introduction to Health care Systems in India	√			
2	Hospital Operations Management	√	√		
3	Health Care Economics, finance & Marketing		√	√	
4	Health Care Information System	√			√
5	Entrepreneurship & Consultancy In Healthcare		√	√	
6	Contemporary Issues	√	√	√	√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communicatio n and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	Learn the foundation of health care structure and its importance in administrative context.	3	3	1	1
CO2	Apply the appropriate managerial techniques to resolve the health care challenges.	3	3	1	1
CO3	Equipping the student to take up entrepreneurship & consultancy work in various facets of healthcare management.	3	3	2	1
CO4	Analyze the role of information system management in health care segment	3	3	3	1